

UBER NCR

RIDE FULFILLMENT EFFICIENCY ANALYSIS

Executive Summary

Business Context: Uber NCR operates one of India's highest-volume ride-hailing markets across Delhi, Gurgaon, Noida, and Faridabad, offering seven vehicle categories. Despite strong booking demand, a critical gap exists between bookings received and rides completed.

The Problem: Analysis of **148,767 bookings in 2024** shows only **92,257 completed rides**, resulting in a **62.01% completion rate**. Nearly **38% of bookings (56,510 rides)** failed to convert, leading to an estimated **₹28.7 million annual revenue leakage**. Customer experience remains strong (Driver Rating: 4.23, Customer Rating: 4.40), confirming the issue is **operational, not experiential**.

Root Cause: Driver-side failures account for **24.99% of bookings**, **3.5x higher** than customer-side failures (7.01%). Additionally, **55% of customer cancellations are driven by driver behaviour**, including not moving to pickup, asking customers to cancel, and non-functional AC. The pattern is consistent across all vehicle types and time periods, indicating a **systemic issue**.

Where and Why It Happens: Geographic analysis across **176 locations** highlights key gaps: **Vinobapuri (54.6%)** and **Akshardham (56.2%)**, both **7–8% below average**, indicating localized supply shortages. Driver cancellation reasons are evenly distributed (~25% each), suggesting broad dissatisfaction. **Vehicle breakdowns (33.5%)** impact incomplete rides, while **wrong address issues (22.5%)** drive customer cancellations, pointing to GPS/product gaps.

Financial Impact: Driver cancellations contribute **₹13.6 million (47.4%)** of total losses, followed by **No Driver Found (₹5.3 million)**. UPI dominates payments (**49.4%**), while Uber Wallet remains underutilized (**13.2%**), indicating an optimization opportunity.

Recommendations & Expected Impact

- Location-based driver incentives → **₹1.5–2.0M recovery (3,000–4,000 rides)**
- Driver accountability program → **₹4.0M recovery (~8,000 rides)**
- GPS & address validation improvements → **₹2.3M recovery**
- Supply strengthening in underserved areas → **₹2.6M recovery (~5,200 rides)**

Together, these initiatives can recover **~20,800 rides**, improve completion rate to **68–70%**, and unlock **~₹10 million in annual revenue**.

1. Business Context

1.1 Company Overview

Uber is a global ride-hailing platform operating in over 70 countries. In India, Uber operates across major metropolitan areas, including the National Capital Region (NCR), which encompasses Delhi, Gurgaon, Noida, Faridabad and surrounding areas. The NCR market represents one of Uber's highest-volume and most competitive markets in the country.

1.2 Vehicle Categories in NCR

Uber NCR serves customers across multiple vehicle segments, catering to different price points and use cases:

- Auto Rickshaw — affordable short-distance urban mobility
- Bike & eBike — fastest option for solo riders
- Go Mini & Go Sedan — economy car options for daily commutes
- Premier Sedan — mid-range comfortable sedan
- Uber XL — large vehicle for groups or extra luggage

1.3 Business Problem Identified

- Despite strong booking volumes, Uber NCR faces a critical operational challenge: a significant share of bookings do not convert into completed rides.
- These failures occur across four categories: **driver-initiated cancellations before pickup, customer cancellations after assignment, no driver found (supply gaps), and incomplete rides that do not reach completion.**
- Each failed booking leads to **revenue loss, poorer customer experience and retention, wasted driver time and resources, and reduced marketplace reliability and trust.**

2. Problem Statement & Hypotheses

2.1 Problem Statement

Uber NCR is experiencing a significant gap between booking demand and successful ride fulfillment. Out of 148,767 total bookings recorded in 2024, only 92,257 rides were completed — representing a Ride Completion Rate of 62.01%. This means 56,510 rides — nearly 38% of all bookings — failed to convert into completed rides, resulting in substantial revenue leakage estimated at ₹28.7 million annually.

Primary Metric: Ride Completion Rate = Completed Rides / Total Bookings
92,257 / 148,767 = 62.01%

2.2 Business Objective

To identify where and why ride bookings fail to convert into completed rides across Uber NCR and recommend practical, data-backed operational improvements to increase the Ride Completion Rate.

2.3 Hypotheses Tested

Sr No	Hypothesis	Result	Finding
1	Certain vehicle types generate high volume but low completion rate	Rejected	All vehicles within 1.1% range — no vehicle-specific issue
2	Some locations face high no-driver-found rates, indicating supply gaps	Confirmed	Vinobapuri (54.6%) and Akshardham (56.2%) show critical gaps
3	Peak evening hours have higher cancellation rates	Rejected	Time of day shows only 0.7% variance — not time-dependent
4	Driver-side failures are the primary cause of booking failures	Confirmed	Driver side = 24.99% vs Customer side = 7.01% (3.5x worse)

3. Data Overview

3.1 Dataset Description

Data Source	Uber NCR Operations Data
Time Period	January 2024 — December 2024
Geography	National Capital Region (NCR), India
Original Rows	150,000
Final Clean Rows	148,767
Total Columns	21 original + 6 helper columns
Unique Pickup Locations	176
Vehicle Categories	7

3.2 Key Metrics Summary

Metric	Value
Total Revenue Generated	₹46,886,024
Average Booking Value	₹508 per ride
Estimated Revenue Lost	₹28,700,000
Average VTAT (Vehicle Arrival Time)	8.46 minutes
Average CTAT (Customer Wait Time)	29.15 minutes
Average Driver Rating	4.23 / 5.0
Average Customer Rating	4.40 / 5.0

3.3 Booking Status Distribution

Booking Status	Count	Percentage	Category
Completed	92,257	62.01%	Success
Cancelled by Driver	26,779	18.00%	Driver Side Failure
Cancelled by Customer	10,424	7.01%	Customer Side Failure
No Driver Found	10,401	6.99%	Driver Side Failure
Incomplete	8,906	5.99%	Partial Completion

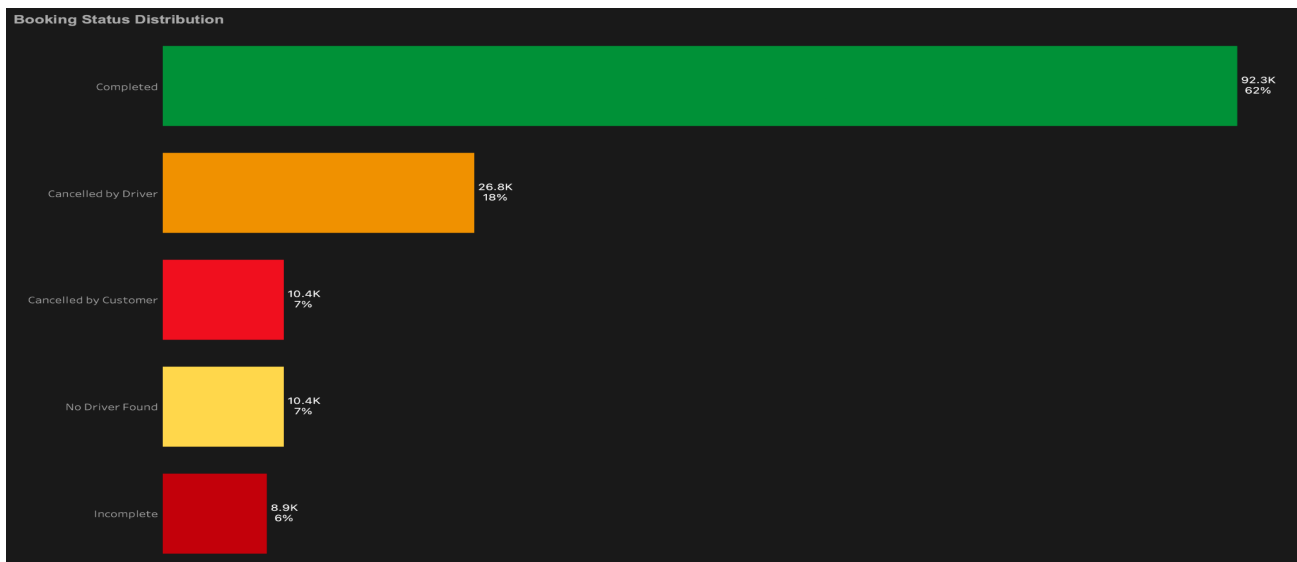


Figure 1: Booking Status Distribution

4. Data Cleaning & Preparation

4.1 Duplicate Treatment

A duplicate check on the **Booking ID** column using `=COUNTIF($C$2:C2, C2)` confirmed it should be unique across all records. The analysis identified **1,233 duplicate IDs**, which showed **inconsistent values across fields** such as Date, Customer ID, and Vehicle Type.

Cleaning Step	Result
Original rows	150,000
Duplicate Booking IDs found	1,233
Method used	Remove Duplicates on Booking ID column
Rows removed	1,233
Final clean rows	148,767

4.2 Missing Value Treatment

A COUNTBLANK analysis was performed across all 21 columns. Null and blank values were identified in 11 of 21 columns. Critically, these missing values were found to be structurally expected and directly linked to Booking Status — not data entry errors.

Column	Nulls	Treatment	Rationale
Avg VTAT	10,401	Retained blank	No Driver Found — driver never arrived
Avg CTAT	47,604	Retained blank	Ride never started — wait time N/A
Booking Value	47,604	Retained blank	No revenue generated for failed rides
Ride Distance	47,604	Retained blank	No ride took place
Driver Ratings	56,510	Retained blank	Rating not applicable without completion
Customer Rating	56,510	Retained blank	Rating not applicable without completion
Payment Method	47,604	Retained blank	No payment made for failed rides
Cancelled by Customer	138,343	Filled with 0	Binary flag — 0 means not cancelled

Column	Nulls	Treatment	Rationale
Cancelled by Driver	121,988	Filled with 0	Binary flag — 0 means not cancelled
Incomplete Rides	139,861	Filled with 0	Binary flag — 0 means not incomplete

4.3 Consistency & Format Validation

All categorical columns were examined using Excel filter dropdowns to identify spelling inconsistencies, case mismatches or trailing spaces. The TRIM() formula was applied to all text columns.

- Booking Status: 5 unique values — no inconsistencies found
- Vehicle Type: 7 unique values — no inconsistencies found
- Payment Method: 5 unique values — no inconsistencies found
- Pickup & Drop Locations: 176 unique values — no inconsistencies found

4.4 Validity Check

MIN() and MAX() functions were applied to all numeric columns to identify outliers or logically impossible values:

Column	Minimum	Maximum	Status
Booking Value (₹)	50	4,277	Valid — no negatives
Ride Distance (km)	1	50	Valid — no zeros
Driver Ratings	3.0	5.0	Valid — within 1-5 range
Customer Rating	3.0	5.0	Valid — within 1-5 range
Avg VTAT (mins)	2	20	Valid — logical range
Avg CTAT (mins)	10	45	Valid — logical range

4.5 Helper Columns Added

Six helper columns were added to support analysis and visualization:

Column Name	Formula	Purpose
Is Completed	=IF(D2="Completed",1,0)	Binary flag for completion rate calculation
Failure Type	IFS formula	Groups failures into business categories
Time Bucket	IFS(HOUR) formula	Group bookings by time of day
Month Name	=TEXT(A2,"MMM")	Enables monthly trend analysis

Column Name	Formula	Purpose
Day of Week	=TEXT(A2,"DDDD")	Enables weekly pattern analysis
Booking Value Bucket	IFS formula	Groups revenue into tiers

5. Key Findings

5.1 Headline Metrics

Analysis of 148,767 Uber NCR bookings for the year 2024 reveals the following findings:

KPI	Value	Implication
Ride Completion Rate	62.01%	38% of demand is being lost
Total Failed Rides	56,510	More than 1 in 3 rides fail
Total Revenue Generated	₹46,886,024	Only from completed rides
Estimated Revenue Lost	₹28,700,000	From 56,510 failed rides
Driver Side Failure Rate	24.99%	Dominant cause of failures
Customer Side Failure Rate	7.01%	Secondary cause
Avg Driver Rating	4.23 / 5	High when rides complete
Avg VTAT	8.46 mins	Driver arrival time
Avg CTAT	29.15 mins	Customer total wait time

5.2 Failure Breakdown

The 56,510 failed rides break down into two distinct categories of failure:

- Driver Side Failures: 37,180 rides (24.99%) — comprising Cancelled by Driver (18.00%) and No Driver Found (6.99%)
- Customer Side Failures: 10,424 rides (7.01%) — Cancelled by Customer only
- Incomplete Rides: 8,906 rides (5.99%) — rides that started but did not complete

Driver-side failures are 3.5 times more frequent than customer-side failures, indicating a supply-side operational problem rather than a demand-side issue. This is the most critical finding of this analysis.



Figure 2: Failure Type Breakdown

5.3 Geographic Analysis

Location analysis across 176 unique pickup points reveals significant performance variance, with certain areas experiencing critically low completion rates — the most actionable finding of this entire analysis.

Rank	Location	Completion Rate	Gap from NCR Avg
1 (Worst)	Vinobapuri	54.6%	-7.4%
2	Akshardham	56.2%	-5.8%
3	Badshahpur	58.0%	-4.0%
4	Chhatarpur	58.1%	-3.9%
5	Pragati Maidan	58.4%	-3.6%
6	Vatika Chowk	58.5%	-3.5%
7	Netaji Subhash Place	58.6%	-3.4%
8	Faridabad Sector 15	58.9%	-3.1%
9	IFFCO Chowk	59.0%	-3.0%
10	Indirapuram	59.1%	-2.9%

Vinobapuri at 54.6% and Akshardham at 56.2% perform nearly 7-8% below the NCR average of 62.01% — highlighting localized supply-side constraints in these specific areas.

These locations likely suffer from insufficient driver availability, low driver incentives and geographic accessibility challenges.



Figure 3: Bottom 10 Locations

5.4 Vehicle Type Analysis

Completion Rate by Vehicle Type reveals a counter-intuitive finding — vehicle type has minimal impact on ride completion:

Vehicle Type	Total Bookings	Completion Rate	vs NCR Average
Go Sedan	26,927	61.5%	-0.5%
Auto	37,114	61.8%	-0.2%
eBike	10,456	62.0%	-0.01%
Premier Sedan	17,977	62.2%	+0.2%
Go Mini	29,541	62.3%	+0.3%
Bike	22,334	62.4%	+0.4%
Uber XL	4,418	62.6%	+0.6%

The variance across all vehicle types is only 1.1% (61.5% to 62.6%). This confirms that vehicle type is NOT a significant driver of failure to complete. The failure distribution across all vehicle categories is remarkably consistent — approximately 62% completed, 18% Driver Churn, 7% Supply Gap, 7% Customer Churn, and 6% Incomplete for every vehicle type. The problem is systemic, not vehicle-specific.

5.5 Time of Day Analysis

Time-based analysis reveals equally minimal variance across all time periods:

Time Period	Completion Rate
Late Night (12-6 am)	62.3%
Morning (6 am-12 pm)	62.2%
Afternoon (12-5 pm)	61.8%
Evening (5-9 pm)	61.9%
Night (9 pm-12 am)	62.5%

Time of day shows only 0.7% variance (61.8% to 62.5%), and shows only 0.9% variance (61.5% to 62.4%). These minimal differences confirm that the completion-rate problem is structural rather than time-dependent. No targeted time-based intervention is warranted.

5.6 Cancellation & Incomplete Ride Analysis

5.6.1 Driver Cancellation Reasons

Driver cancellations (26,779 rides — 18% of total bookings) represent the single largest failure category. The breakdown of reasons reveals a critical finding:

Driver Cancellation Reason	Count	Percentage
Customer-related issue	6,780	25.3%
The customer was coughing/sick	6,697	25.0%
Personal & Car-related issues	6,676	24.9%
Exceeding passenger capacity	6,626	24.7%
TOTAL	26,779	100.0%

All four cancellation reasons are remarkably equal at approximately 25% each. This indicates no single dominant cause but rather broad driver dissatisfaction requiring multiple targeted interventions rather than one specific fix.

5.6.2 Customer Cancellation Reasons

Customer cancellations (10,424 rides — 7% of total bookings) show a more distributed pattern:

Customer Cancellation Reason	Count	Percentage
Wrong Address	2,345	22.5%

Customer Cancellation Reason	Count	Percentage
Change of plans	2,337	22.4%
The driver is not moving towards the pickup	2,319	22.2%
The driver asked to cancel	2,279	21.9%
AC is not working	1,144	11.0%
TOTAL	10,424	100.0%

A key insight: **55.1% of customer cancellations are driven by driver behaviour**—including not moving toward pickup (22.2%), asking customers to cancel (21.9%), and non-functional AC (11.0%). Improving driver accountability would reduce both driver- and customer-side cancellations.

5.6.3 Incomplete Rides Reasons

Vehicle Breakdown (33.5%) highlights a significant fleet maintenance issue. Customer Demand (33.8%) suggests customers stopping rides midway — potentially related to pricing dissatisfaction or app experience issues.

Incomplete Ride Reason	Count	Percentage
Customer Demand	3,011	33.8%
Vehicle Breakdown	2,981	33.5%
Other Issue	2,914	32.7%
TOTAL	8,906	100.0%

5.7 Revenue Impact Analysis

The financial impact of ride failures is substantial. Using the average booking value of ₹508 per completed ride as the basis for estimation:

Failure Category	Failed Rides	Est. Revenue Lost	% of Total Loss
Cancelled by Driver	26,779	₹13,603,732	47.4%
No Driver Found	10,401	₹5,283,708	18.4%
Cancelled by Customer	10,424	₹5,295,392	18.4%
Incomplete Rides	8,906	₹4,524,248	15.8%
TOTAL	56,510	₹28,707,080	100.0%

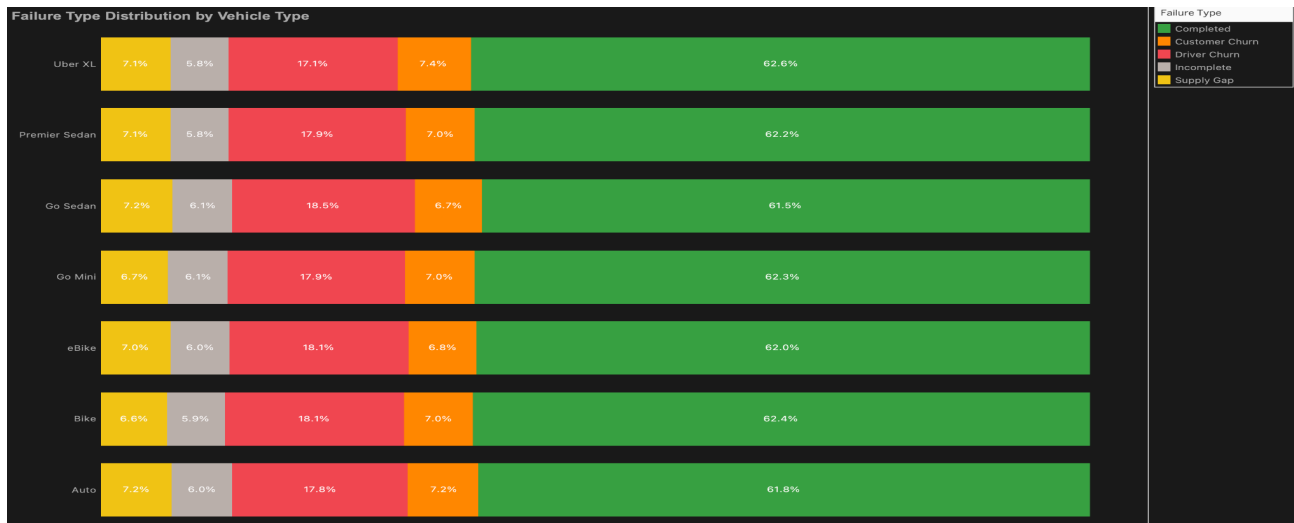


Figure 4: Failure Type Distribution by Vehicle Type

Driver cancellations cause an annual loss of ₹13.6M (~50% of leakage), making them the top operational priority.

5.8 Payment Method Analysis

Payment Method Distribution (Completed Rides Only):

Method	Count	%	Avg Value
UPI	45,558	49.4%	₹508 ▾
Cash	25,138	27.3%	₹498 ▾
Uber Wallet	12,184	13.2%	₹512 ▾
Credit Card	10,110	11.0%	₹519 ▾
Debit Card	8,173	8.9%	₹505 ▾
TOTAL	92,257	100.0%	₹508 ▾

Key Insights

- UPI dominates (49.4%), accounting for nearly half of all rides, ensuring reliability and promoting usage
- Cash remains significant (27.3%), indicating strong dependency, removing it risks ~25K rides
- Uber Wallet is underutilized (13.2%), despite the cost advantages and opportunity via cashback incentives
- Cards (19.9%) indicate a premium segment, likely with higher lifetime value

6. Recommendations

Sr No	Action	Priority	Key Actions	Impact
1	Location Specific Driver Allocation	High	Incentives in the bottom 10 locations, surge pricing in low-coverage areas, minimum driver availability targets, and demand-based driver pre-positioning	Recover 3,000–4,000 rides annually; ₹1.5–2.0M revenue
2	Driver Accountability & Quality Program	High	Cancellation penalties; driver training, monitor pickup movement, enforce vehicle maintenance (33.5% breakdown), minimum rating threshold	Reduce cancellations by 30%; recover ~8,000 rides; ₹4.0M revenue
3	GPS & Address Validation Enhancement	Medium	Improve GPS & validation; confirm pickup feature; real-time tracking; in-app messaging	Recover 4,600+ rides annually; ₹2.3M revenue
4	Supply-Side Strengthening	Medium	Driver recruitment; joining & performance incentives; dynamic pricing; aggregator partnerships	Recover ~5,200 rides annually; ₹2.6M revenue

7. Conclusion & Limitations

7.1 Summary of Findings

This analysis of 148,767 Uber NCR bookings for 2024 reveals that the platform's primary challenge is a structural supply-side problem. The Ride Completion Rate of 62.01% indicates that 38% of customer demand is being lost — representing an estimated ₹28.7 million in annual revenue leakage.

The five most important conclusions from this analysis are:

- Driver side failures (24.99%) are 3.5x worse than customer side failures (7.01%) — confirming this is primarily a supply problem
- The problem is NOT concentrated in any specific vehicle type or time period — it is systemic across all segments
- The problem IS geographically concentrated — 10 locations, particularly Vinobapuri (54.6%) and Akshardham (56.2%), show critically low completion rates
- Driver cancellation reasons are evenly distributed across four categories, suggesting broad dissatisfaction requiring a comprehensive multi-pronged intervention
- 55% of customer cancellations are also caused by driver behaviour — making driver accountability the single highest-leverage intervention point

7.2 Limitations

- Analysis is based on 2024 data only — seasonal trends across multiple years not considered
- Revenue loss is estimated using average booking value — actual values would vary by ride distance and vehicle type
- External factors such as weather events, competitor activity and regulatory changes are not accounted for
- Driver supply constraints and operational costs of implementing recommendations have not been modelled
- The customer lifetime value impact of failed rides has not been quantified.